

Continuous Random Variables

ANSWER KEY



Verifying a valid probability density function

- 1 A continuous random variable X has probability density function $f(x) = \frac{1}{4}$ for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise. Verify that this is a valid pdf.
Since $f(x) = \frac{1}{4} \geq 0$ throughout, and $\int_0^4 \frac{1}{4} dx = \frac{1}{4}(4) = 1$, the total area under f equals 1, confirming a valid pdf (the uniform distribution on $[0, 4]$).
- 2 Determine the value of c such that $f(x) = c$ for $2 \leq x \leq 6$, and $f(x) = 0$ otherwise, is a valid pdf.
 $\int_2^6 c dx = 4c = 1 \Rightarrow c = \frac{1}{4}$.

Finding an unknown constant in a pdf

- 3 A continuous random variable X has pdf $f(x) = kx$ for $0 \leq x \leq 5$, and $f(x) = 0$ otherwise. Find k .
 $\int_0^5 kx dx = k \left[\frac{x^2}{2} \right]_0^5 = \frac{25k}{2} = 1 \Rightarrow k = \frac{2}{25}$.
- 4 A continuous random variable X has pdf $f(x) = k(4 - x)$ for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise. Find k .
 $\int_0^4 k(4 - x) dx = k \left[4x - \frac{x^2}{2} \right]_0^4 = k(16 - 8) = 8k = 1 \Rightarrow k = \frac{1}{8}$.
- 5 A continuous random variable X has pdf $f(x) = kx^2$ for $0 \leq x \leq 3$, and $f(x) = 0$ otherwise. Find k .
 $\int_0^3 kx^2 dx = k \left[\frac{x^3}{3} \right]_0^3 = 9k = 1 \Rightarrow k = \frac{1}{9}$.

Finding probabilities by integrating the pdf

- 6 Using $f(x) = \frac{2}{25}x$ for $0 \leq x \leq 5$, find $P(1 < X < 2)$.
 $\int_1^2 \frac{2}{25}x dx = \frac{2}{25} \left[\frac{x^2}{2} \right]_1^2 = \frac{2}{25}(2) = \frac{4}{25} = 0.16$.
- 7 Using the same pdf, find $P(2 < X < 4)$.
 $\int_2^4 \frac{2}{25}x dx = \frac{2}{25} \left[\frac{x^2}{2} \right]_2^4 = \frac{2}{25}(8 - 2) = \frac{12}{25} = 0.48$.
- 8 Using $f(x) = \frac{1}{8}(4 - x)$ for $0 \leq x \leq 4$, find $P(2 < X < 4)$.
 $\int_2^4 \frac{1}{8}(4 - x) dx = \frac{1}{8} \left[4x - \frac{x^2}{2} \right]_2^4 = \frac{1}{8}[(16 - 8) - (8 - 2)] = \frac{1}{8}(2) = 0.25$.
- 9 Using $f(x) = \frac{1}{9}x^2$ for $0 \leq x \leq 3$, find $P(X < 1)$.
 $\int_0^1 \frac{1}{9}x^2 dx = \frac{1}{9} \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{9} \cdot \frac{1}{3} = \frac{1}{27} \approx 0.037$.

The cumulative distribution function

- 10 For the pdf $f(x) = \frac{2}{25}x$ on $0 \leq x \leq 5$, find the cumulative distribution function $F(x)$.
 $F(x) = \int_0^x \frac{2}{25}t dt = \frac{2}{25} \cdot \frac{x^2}{2} = \frac{x^2}{25}$, for $0 \leq x \leq 5$.
- 11 Using $F(x) = \frac{x^2}{25}$, find $P(X \leq 3)$ and hence $P(X > 3)$.
 $F(3) = \frac{9}{25} = 0.36$. $P(X > 3) = 1 - 0.36 = 0.64$.
- 12 For the pdf $f(x) = \frac{1}{8}(4-x)$ on $0 \leq x \leq 4$, find the cumulative distribution function $F(x)$.
 $F(x) = \int_0^x \frac{1}{8}(4-t) dt = \frac{1}{8} \left(4x - \frac{x^2}{2} \right) = \frac{x}{2} - \frac{x^2}{16}$. Checking $F(4) = 2 - 1 = 1$ confirms the total probability is correctly 1.
- 13 For the pdf $f(x) = \frac{1}{9}x^2$ on $0 \leq x \leq 3$, find the cumulative distribution function $F(x)$.
 $F(x) = \int_0^x \frac{1}{9}t^2 dt = \frac{1}{9} \cdot \frac{x^3}{3} = \frac{x^3}{27}$.

Using the cdf to find probabilities

- 14 Given $F(x) = \frac{x^2}{25}$ for $0 \leq x \leq 5$, find $P(1 \leq X \leq 4)$.
 $F(4) - F(1) = \frac{16}{25} - \frac{1}{25} = \frac{15}{25} = 0.6$.
- 15 Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$, find $P(1 \leq X \leq 3)$.
 $F(3) = \frac{3}{2} - \frac{9}{16} = \frac{15}{16}$. $F(1) = \frac{1}{2} - \frac{1}{16} = \frac{7}{16}$. $P(1 \leq X \leq 3) = \frac{15}{16} - \frac{7}{16} = \frac{8}{16} = 0.5$.
- 16 Using $F(x) = \frac{x^3}{27}$ for $0 \leq x \leq 3$, find $P(1 \leq X \leq 2)$.
 $F(2) - F(1) = \frac{8}{27} - \frac{1}{27} = \frac{7}{27} \approx 0.259$.

Expectation of a continuous random variable

- 17 Using $f(x) = \frac{2}{25}x$ for $0 \leq x \leq 5$, find $E(X)$, using $E(X) = \int_0^5 xf(x) dx$.
 $E(X) = \int_0^5 x \cdot \frac{2}{25}x dx = \frac{2}{25} \left[\frac{x^3}{3} \right]_0^5 = \frac{2}{25} \cdot \frac{125}{3} = \frac{10}{3} \approx 3.333$.
- 18 Using the uniform pdf $f(x) = \frac{1}{4}$ for $0 \leq x \leq 4$, find $E(X)$, and verify it matches the midpoint of the interval.
 $E(X) = \int_0^4 x \cdot \frac{1}{4} dx = \frac{1}{4} \left[\frac{x^2}{2} \right]_0^4 = \frac{1}{4} (8) = 2$, matching the midpoint of $[0, 4]$, as expected for a uniform distribution.
- 19 Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, find $E(X)$.
 $E(X) = \int_0^3 x \cdot \frac{1}{9}x^2 dx = \frac{1}{9} \left[\frac{x^4}{4} \right]_0^3 = \frac{1}{9} \cdot \frac{81}{4} = \frac{9}{4} = 2.25$.

Variance of a continuous random variable

- 20 Using $f(x) = \frac{2}{25}x$ on $[0, 5]$ and $E(X) = \frac{10}{3}$ (found earlier), find $\text{Var}(X)$, using $\text{Var}(X) = E(X^2) - [E(X)]^2$.
 $E(X^2) = \int_0^5 x^2 \cdot \frac{2}{25}x dx = \frac{2}{25} \left[\frac{x^4}{4} \right]_0^5 = \frac{2}{25} \cdot \frac{625}{4} = 12.5$.
 $\text{Var}(X) = 12.5 - \left(\frac{10}{3} \right)^2 = 12.5 - \frac{100}{9} \approx 1.389$.

- 21 Using the uniform pdf $f(x) = \frac{1}{4}$ on $[0, 4]$ and $F(x) = \frac{x}{4}$ (found earlier), find $\text{Var}(X)$.
 $E(X^2) = \int_0^4 x^2 \cdot \frac{1}{4} dx = \frac{1}{4} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{4} \cdot \frac{64}{3} = \frac{16}{3}$. $\text{Var}(X) = \frac{16}{3} - 2^2 = \frac{16}{3} - \frac{12}{3} = \frac{4}{3} \approx 1.333$.
- 22 Using $f(x) = \frac{1}{\sqrt{2}}$ on $[0, 2]$ and $F(x) = \frac{x^2}{2}$ (found earlier), find $\text{Var}(X)$.
 $E(X^2) = \int_0^2 x^2 \cdot \frac{1}{\sqrt{2}} dx = \frac{1}{\sqrt{2}} \left[\frac{x^3}{3} \right]_0^2 = \frac{1}{\sqrt{2}} \cdot \frac{8}{3} = \frac{4\sqrt{2}}{3} \approx 5.4$.
 $\text{Var}(X) = 5.4 - \left(\frac{9}{4}\right)^2 = 5.4 - 5.0625 = 0.3375$.

Median and quartiles

- 23 Using the cdf $F(x) = \frac{x^2}{25}$ for $0 \leq x \leq 5$, find the median m of X , solving $F(m) = 0.5$.
 $\frac{m^2}{25} = 0.5 \Rightarrow m^2 = 12.5 \Rightarrow m = \sqrt{12.5} \approx 3.536$.
- 24 Using the same cdf, find the lower quartile Q_1 , solving $F(Q_1) = 0.25$.
 $\frac{Q_1^2}{25} = 0.25 \Rightarrow Q_1^2 = 6.25 \Rightarrow Q_1 = 2.5$.
- 25 Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$ for $0 \leq x \leq 4$, find the upper quartile Q_3 , solving $F(Q_3) = 0.75$.
 $\frac{x}{2} - \frac{x^2}{16} = 0.75$. Multiplying by 16: $8x - x^2 = 12 \Rightarrow x^2 - 8x + 12 = 0 \Rightarrow (x-2)(x-6) = 0$. Since the domain is $[0, 4]$, $x = 6$ is rejected, so $Q_3 = 2$.
- 26 Using the cdf $F(x) = \frac{x^3}{27}$ for $0 \leq x \leq 3$, find the median m of X , to three decimal places.
 $\frac{m^3}{27} = 0.5 \Rightarrow m^3 = 13.5 \Rightarrow m = 13.5^{1/3} \approx 2.381$.

A well-designed word problem: waiting time at a bus stop

- 27 This question has four parts. The waiting time X (minutes) for a bus is modeled by $f(x) = \frac{3}{64}x^2$ for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise. (a) Verify this is a valid pdf. (b) Find the cdf $F(x)$, and use it to find the probability a passenger waits more than 3 minutes. (c) Find $E(X)$, the mean waiting time. (d) Find the median waiting time, to three decimal places.
- (a) $\int_0^4 \frac{3}{64}x^2 dx = \frac{3}{64} \left[\frac{x^3}{3} \right]_0^4 = \frac{3}{64} \cdot \frac{64}{3} = 1$, and $f(x) \geq 0$ throughout, confirming a valid pdf. (b)
 $F(x) = \int_0^x \frac{3}{64}t^2 dt = \frac{3}{64} \left[\frac{t^3}{3} \right]_0^x = \frac{3}{64} \cdot \frac{x^3}{3} = \frac{x^3}{64}$. $F(3) = \frac{27}{64} \approx 0.422$. (c)
 $E(X) = \int_0^4 x \cdot \frac{3}{64}x^2 dx = \frac{3}{64} \left[\frac{x^4}{4} \right]_0^4 = \frac{3}{64} \cdot \frac{256}{4} = 3$ minutes. (d) Solving $F(m) = 0.5$:
 $\frac{m^3}{64} = 0.5 \Rightarrow m^3 = 32 \Rightarrow m = 32^{1/3} \approx 3.175$ minutes.